Capital Budgeting

- Net Present Value
- Internal Rate of Return
- Profitability Index
- Payback Period

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Capital budgeting Techniques

Investment decisions are generally called capital budgeting decisions

How do you decide - whether you should invest in **Project A** or **Project B** ?

Criteria

- NPV What is the value of the return on investment.
- IRR % of Return on investment.
- Profitability Index Profit ratio for every dollar spent.
- Payback period Time to recover your investments, A Break even point.

Opportunity cost

1. Invest in a Bank - Bank Annual Returns: 8%
2. Invest in a Project - Project Annual Returns : 10%

What is the opportunity cost if you invest in the Bank?

What is the opportunity cost if you invest in the Project?

So, where should you invest your money?

Cash Flows

1. Discounted Cash Flow
2. Non Discounted Cash Flow

Discounted cash flow

1. Discounted Cash flow is nothing but the Opportunity Cost.

2. Rate of return that an organization could have earned on the investment, if not invested in the current project.

3. In other words, it's the rate of return that an organization is willing to loose in an expectation to earn more by investing in this project.

4. It is the lost opportunity on the capital that is being invested in the projects.

5. Other names for Discounted cash flow

    - Opportunity Cost of Capital
    - Cost of Capital

Discounted cash flow

Let’s say If an organization earns 10% interest per annum on its capital by putting the money in bank instead of investing in the project.

What is the opportunity cost of the capital ?

10% - You are correct.

Since it is the minimum that an organization could have earned if invested the money in the bank.

Non Discounted cash flow

- In Non discounted cash flow, the interest rate, opportunity cost or Discounted cash flow is not taken in to consideration.

Consideration for Capital budgeting

Discounted Cash flow

- NPV - Net present Value
- IRR - Internal Rate of Return
- PI – Profitability Index

Non-Discounted Cash flow

- Payback period (Payback period is usually calculated considering the Non discounted cash flow.

Time Value of Money

A Dollar sitting in your wallet is worth more today than the same dollar tomorrow.

Can you tell Why?

- Depreciation.
- Money grows over time when it earns interest.

So, the time value of money brings us to the concept of

- Future Value of Money.
- Present Value of Money.

Relationship between FV & PV

- Future Value of Money (let's call it FV)
- Present Value of Money (let's call it PV)

$$FV = PV\,(1 + K)^n$$

FV = Future Value
PV = Present Value
K = Discounted Rate in %
n = Number of Years

Relationship between FV & PV

$$FV = PV\,(1 + K)^n \quad \Rightarrow \quad PV = \frac{FV}{(1 + K)^n}$$

Calculate the Future Value

Example: If $ 100 dollars is invested in a bank today may earn 8% per year. what is the future value of the $ 100 dollars for 1st, 5th and 15th year?

$$FV = PV\,(1 + k)^n$$

FV = Future Value
PV = Present Value
K = Discounted Rate
n = Number of Years

PV = 100
n = 1
n = 5
n = 15
K = 8%
FV = ? When n =1, n=5, n=15

K =	8%	=	8/100	=	0.08
(1+K)=	(1+0.08)	=	1.08	=	1.08

After 1 year(n=1): FV = 100 X $(1.08)^1$ = 100*1.08 = **$108**
After 5 years (n=5): FV = 100 X $(1.08)^5$ = 100*1.08*1.08*1.08*1.08*1.08 = **$146.93**
After 15 years (n=15): FV = 100 X $(1.08)^{15}$ = **$317.22**

Calculate the Present Value

Example

If $ 100 dollars is to be received after 1 year, what is the present value of $100 dollars today?

If $ 100 dollars is to be received after 5 years, what is the present value of $100 dollars today?

If $ 100 dollars is to be received after 15 years, what is the present value of $100 dollars today?

Note: Discounted rate is 8% per year.

$$PV = \frac{FV}{(1 + K)^n}$$

FV = 100
n = 1
n = 5
n = 15
K = 8%
PV = ? for n =1, n=5, n=15

Year	FV	PV
1	100	$\frac{100}{(1.08)^1}$
5	100	$\frac{100}{(1.08)^5}$
15	100	$\frac{100}{(1.08)^{15}}$

K =	8%	=	8/100	=	0.08
(1+K)=	(1+0.08)	=	1.08	=	1.08

The Present value of $ 100 to be received after 1 year is **$93** dollars today.

The Present value of $ 100 to be received after 5 years is **$68** dollars today.

The Present value of $ 100 to be received after 15 year is **$32** dollars today.

A quick recap so far…

So far we learned about

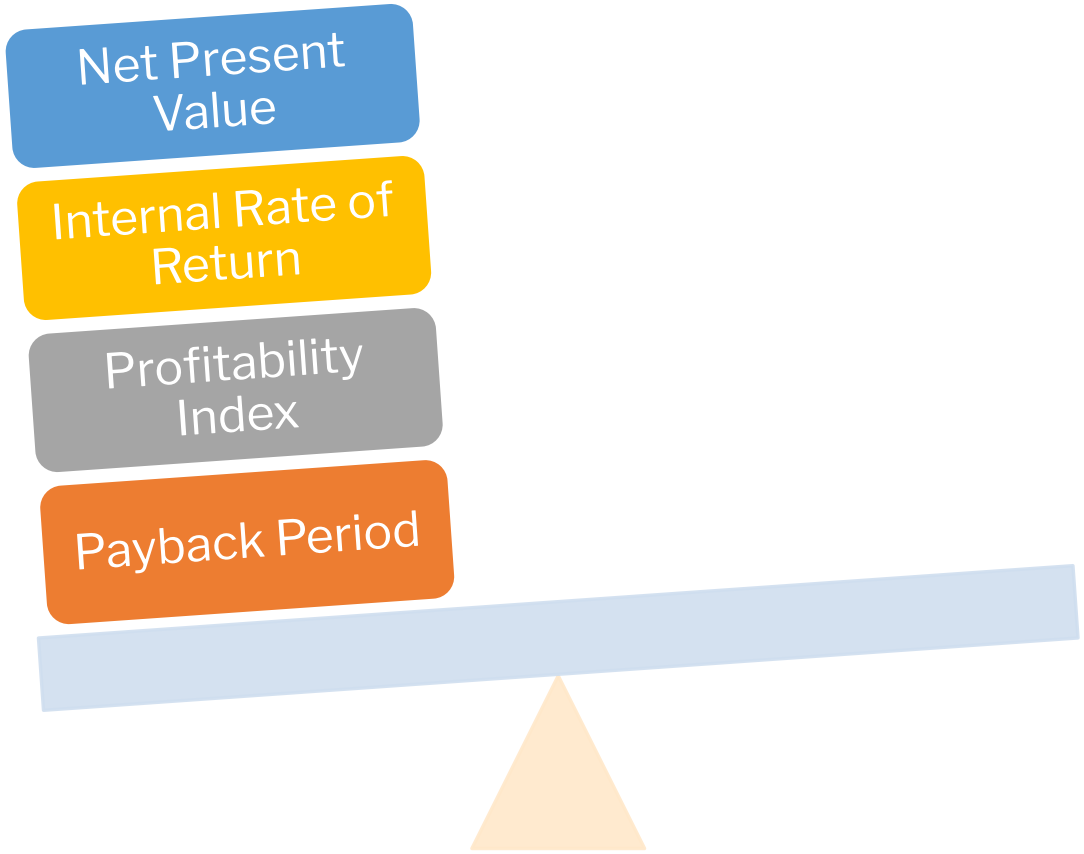
- Discounted Cash flow (Opportunity cost)
- Non Discounted Cash flow
- Future Value
- Present Value

Formula of FV i.e., $FV = PV\,(1 + K)^n$

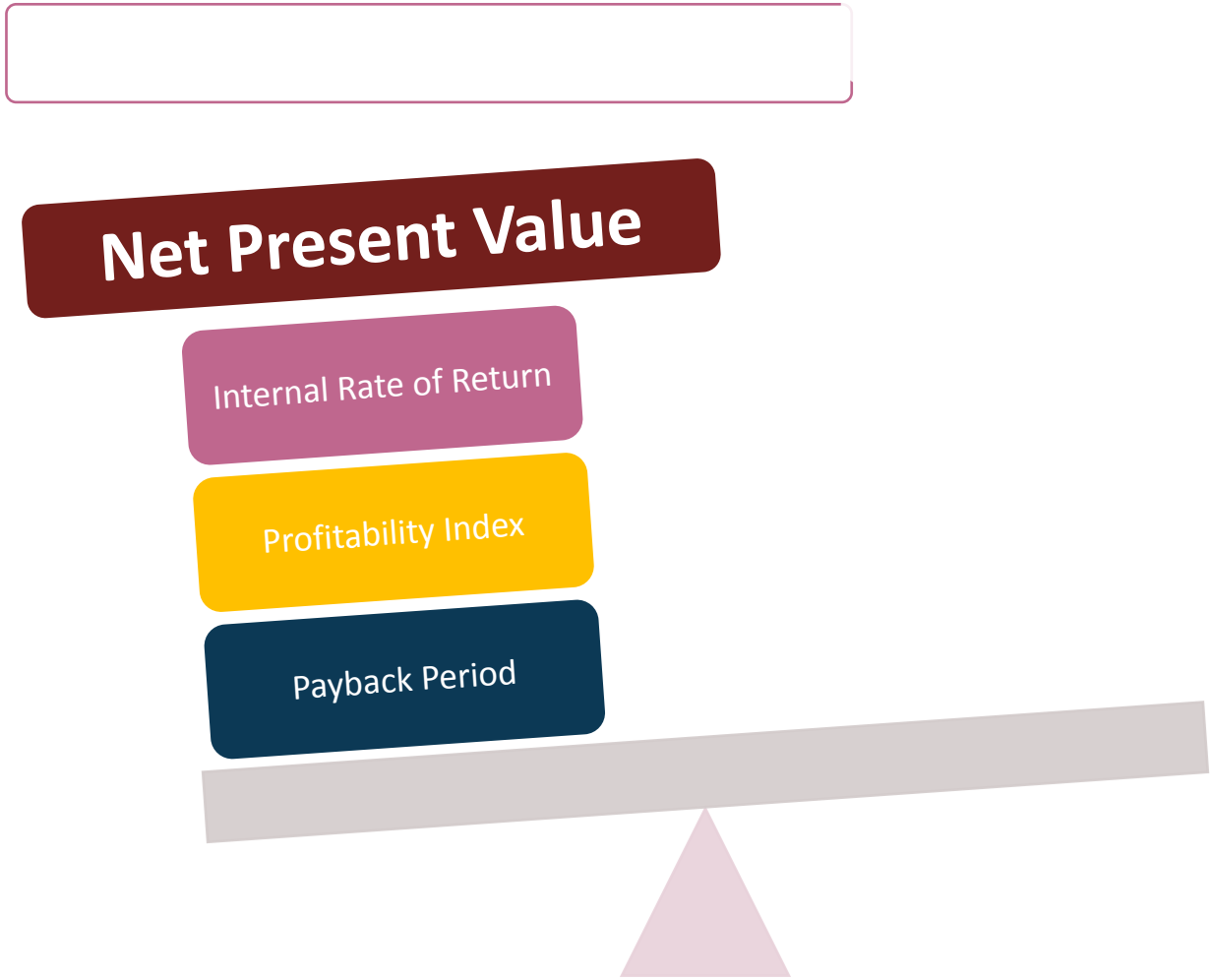
Formula of PV i.e., $PV = \dfrac{FV}{(1 + K)^n}$

FV = Future Value
PV = Present Value
K = Discounted Rate
n = Number of Years

Let’s now do the calculations



Net Present Value



Net Present Value

Net Present Value (NPV) = “Present Value of all cash inflows – Present Value of all cash outflow”

Example: Salary Slip

Net Salary = Gross Salary – Deductions

Similarly, The **Present Value of all cash inflows** is the **Gross Present Value**

and if you **deduct cash outflows** it becomes your **Net Present Value.**

Net Present Value

Criteria

If NPV > 0 (NPV is +ve, Accept the Project)

If NPV < 0 (NPV is –ve, Reject the Project)

If NPV = 0 (Accept the Project, considering other non tangible benefits)

Greater the NPV, Better the Prospects

Example1: Calculating NPV

A sum of **$ 400,000** dollars invested today in an IT project may give a series of below cash inflows in future:

$ 70,000 in year 1
$ 120,000 in year 2
$ 140,000 in year 3
$ 140,000 in year 4
$ 40,000 in year 5

If Opportunity cost of capital is 8% per annum, then should we accept or reject the project?

$$PV = FV \times \frac{1}{(1 + k)^n}$$

Solution:
Step 1: Calculate the PV value of year 1, year2, year3, year4, and year5
Step 2: Sum up the PV of all years
Step3: NPV = Present value of all cash inflows – Present value of all cash outflow.
Step 4: If NPV is positive, Accept the project, if not Reject the project.

Example1: Calculating NPV

Formula for calculating Present value = $PV = FV \times \frac{1}{(1+k)^n}$

Step 1: Calculating Present Value of all the Years						
PV for Year 1	=	$70000/(1.08)^1$	=	70000/1.08	=	64814.8
PV for Year 2	=	$120000/(1.08)^2$	=	120000/1.17	=	102880.7
PV for Year 3	=	$140000/(1.08)^3$	=	140000/1.26	=	111136.5
PV for Year 4	=	$140000/(1.08)^4$	=	140000/1.47	=	102904.2
PV for Year 5	=	$40000/(1.08)^5$	=	40000/1.59	=	27223.3
Cash inflow for all the Present Values					=	408959.5

Cash Inflow of all Present Values is : $ 408,959
Present value of Cash outflow is : $400,000

Net Present Value PV of Cash inflows – PV of Cash Outflows
 ($408959 – $400000) = $8959 dollars.

Since NPV is positive, (i.e., $8959, This project can be accepted)

Example2: Calculating NPV

Same example: Calculating NPV however with Discount rate or Opportunity cost of capital at 15%
A sum of $ 400,000 dollars invested today in an IT project may give a series of below cash inflows in future:

$ 70,000 in year 1
$ 120,000 in year 2
$ 140,000 in year 3
$ 140,000 in year 4
$ 40,000 in year 5

If Opportunity cost of capital is 15% per annum, then should we accept or reject the project?

Solution: Calculating NPV
Step 1: Calculate the PV value of year 1, year2, year3, year4, and year5
Step 2: Sum up the PV of all years
Step3: NPV = Present value of all cash inflows – Present value of all cash outflow.
Step 4: If NPV is positive, Accept the project, if not Reject the project.

Example2: Calculating NPV

Formula for calculating Present value = $PV = FV \times \frac{1}{(1+k)^n}$

Step 1: Calculating Present Value of all the Years						
PV for Year 1	=	$70000/(1.15)^1$	=	70000/1.15	=	60869.6
PV for Year 2	=	$120000/(1.15)^2$	=	120000/1.32	=	90737.2
PV for Year 3	=	$140000/(1.15)^3$	=	140000/1.52	=	92052.3
PV for Year 4	=	$140000/(1.15)^4$	=	140000/1.75	=	80045.5
PV for Year 5	=	$40000/(1.15)^5$	=	40000/2.01	=	19887.1
Cash inflow for all the Present Values					=	343591.6

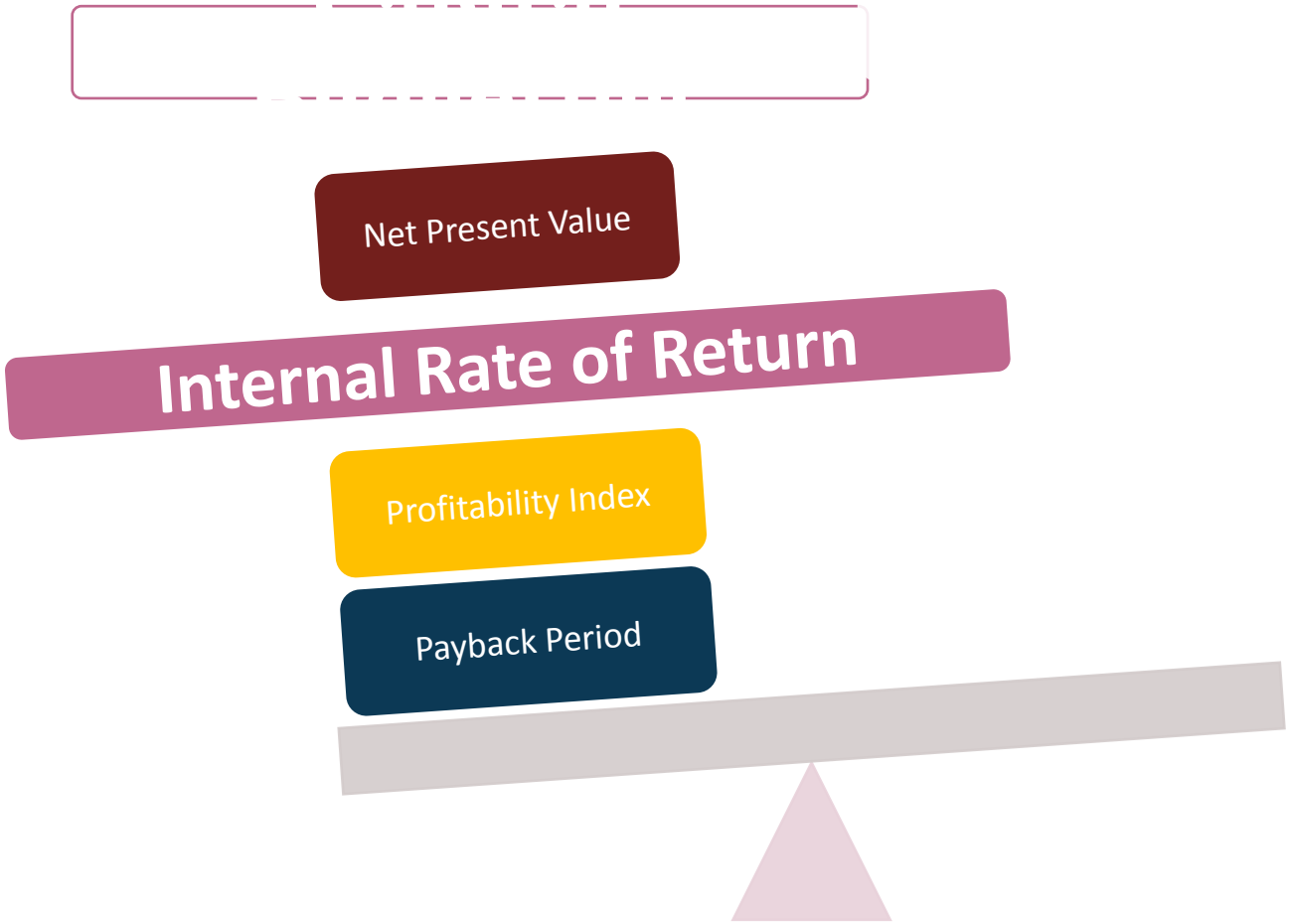
N.B: Though we have the same inflow & outflow of cash as in the previous example, the NPV value changed with the change in the Discount rate of interest.
Therefore, NPV is very much dependent on the Discount rate or in other words the opportunity cost of the capital value.

Cash Inflow of all Present Values is : $ 343591
Present value of Cash outflow is : $400,000

Net Present Value PV of Cash inflows – PV of Cash Outflows
 ($343591 – $400000) = $ -56408 dollars.
(negative 56408 dollars)

Since NPV is Negative, (i.e., - $56408, This project should be rejected)

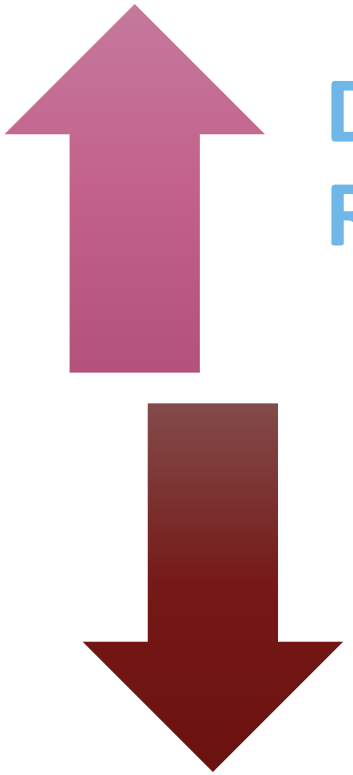
Internal Rate of Return



Internal Rate of Return (IRR)

- IRR □ % of Return on investment.
- To put it simple: It is the percentage of Return of your investment.
- How do we calculate IRR?
  - If you remember from NPV example, we mentioned that the NPV is dependent on Discounted rate
  - If we increase the discount rate the NPV value decreases
  - We need to increase /decrease the discount rate to a level where NPV becomes zero
  - The discount rate at which NPV becomes zero is infact the Internal rate of return
- In other words, IRR is the opportunity cost at which the NPV becomes Zero.

IRR



Let's say at 8%, Discount rate, the NPV is 5000

And

Let's say at 12%, Discount rate, the NPV is 0

Then 12% is the return, we are getting from the project

And

This 12% is called the IRR.

Internal Rate of Return (IRR)

Note:

- For Constant rate of Cash inflow for every year, Internal Rate of Return can be calculated with the help of a formula.
- For Uneven rate of Cash inflows for every year, IRR can be calculated by little trail & error adjustments.

Accept the project when Internal rate of return > Discount rate or Opportunity cost of capital.

Reject the project when Internal rate of return < Discount rate or Opportunity cost of capital.

May accept the project when Internal rate of return = Discount rate or Opportunity cost of capital.

Relationship between IRR, Discount rate and NPV

If IRR > Discount rate or Opportunity cost of capital The NPV is always Positive.

If IRR < Discount rate or Opportunity cost of capital The NPV is always Negative.

If IRR = Discount rate or Opportunity cost of capital The NPV is Zero.

Note: As long as the NPV is Positive, the project is financially viable.
The moment that NPV becomes Negative, the Project is NOT financially viable.

Calculating Internal Rate of Return

Example:

The cost of a project is $1000. It has a time horizon of 5 years and the expected year wise incremental cash flows are:

Year 1 : $200
Year 2 : $300
Year 3 : $300
Year 4 : $400
Year 5 : $500

Compute IRR of the project. If opportunity cost of Capital is 12%, And tell us, should we accept the project?

Solution:
Step 1: Take “K” as 12% and calculate NPV value.
Step 2: If NPV < 0 then Project is NOT financially viable at 12% discount rate.
Step 3: If NPV > 0 then Project is financially viable at 12% however we need to know the actual IRR value, so we need to increase the K value to and calculate the NPV, continue it till you reach a point where the NPV becomes zero or close to zero.

Step 4: The “K” value at which NPV becomes Zero or “Near Zero” is the actual IRR (Internal Rate of Return)

Calculating Internal Rate of Return

K (Discount rate)	12
K%	0.12
(1+K)	1.12

IRR using Variable Discount Rate				
Year (n)	FV	$(1+K)^n$		$PV = FV \times \frac{1}{(1+k)^n}$
1	200	$(1+K)^1$	1.12	**178.57**
2	300	$(1+K)^2$	1.254	**239.16**
3	300	$(1+K)^3$	1.4049	**213.53**
4	400	$(1+K)^4$	1.57352	**254.21**
5	500	$(1+K)^5$	1.762342	**283.71**
Total Cash Inflow				**1169.18**

NPV = Cash Inflow - Cash Outflow		
Cash Inflow	=	**1169.18**
Cash Outflow	=	**1000**
NPV	=	169.18

IRR

Example:

The cost of a project is **$1000**.
It has a time horizon of 5 years and the expected year wise incremental cash flows are:

Year 1 : $200
Year 2 : $300
Year 3 : $300
Year 4 : $400
Year 5 : $500

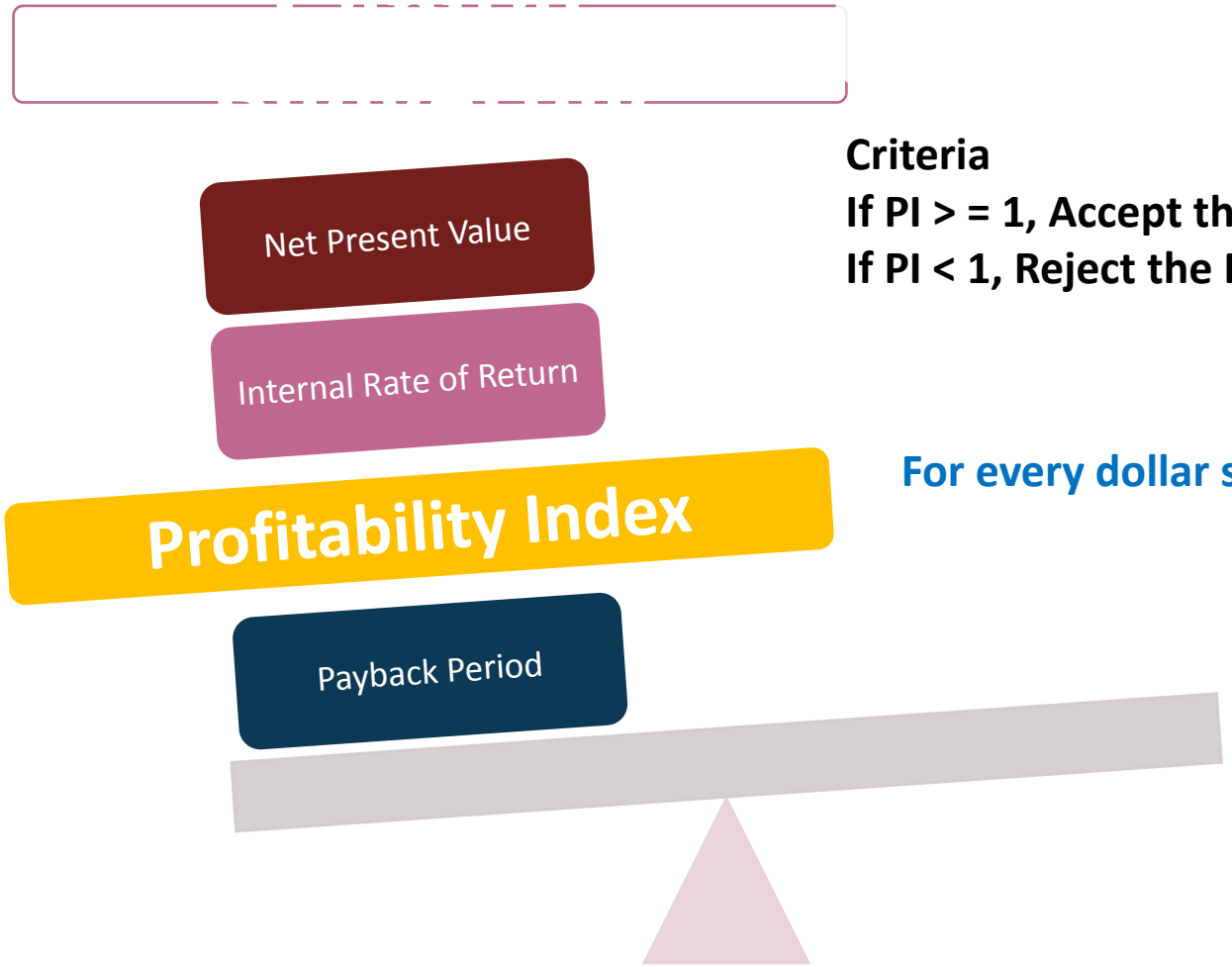
Compute IRR of the project.
If opportunity cost of Capital is 12%,
And tell us, should we accept the project?

At Discount Rate of 12%, the NPV is 169 (positive)

At Discount Rate of 17.7%, the NPV is 0 (Zero), therefore the IRR is 17.7%

Since IRR > Discount rate, Project can be accepted

Profitability Index (PI)



Criteria
If PI > = 1, Accept the Project.
If PI < 1, Reject the Project.

For every dollar spent, how much are we getting back

Profitability Index

A sum of $ 25,000 invested today in a project may give a series of cash inflows in future as described below:

- $ 5000 in year 1
- $ 9000 in year 2
- $ 10,000 in each of year 3
- $ 10,000 in each of year 4
- $ 3000 in year 5

If the required rate of return is 12% pa, what is the Profitability Index?

PI

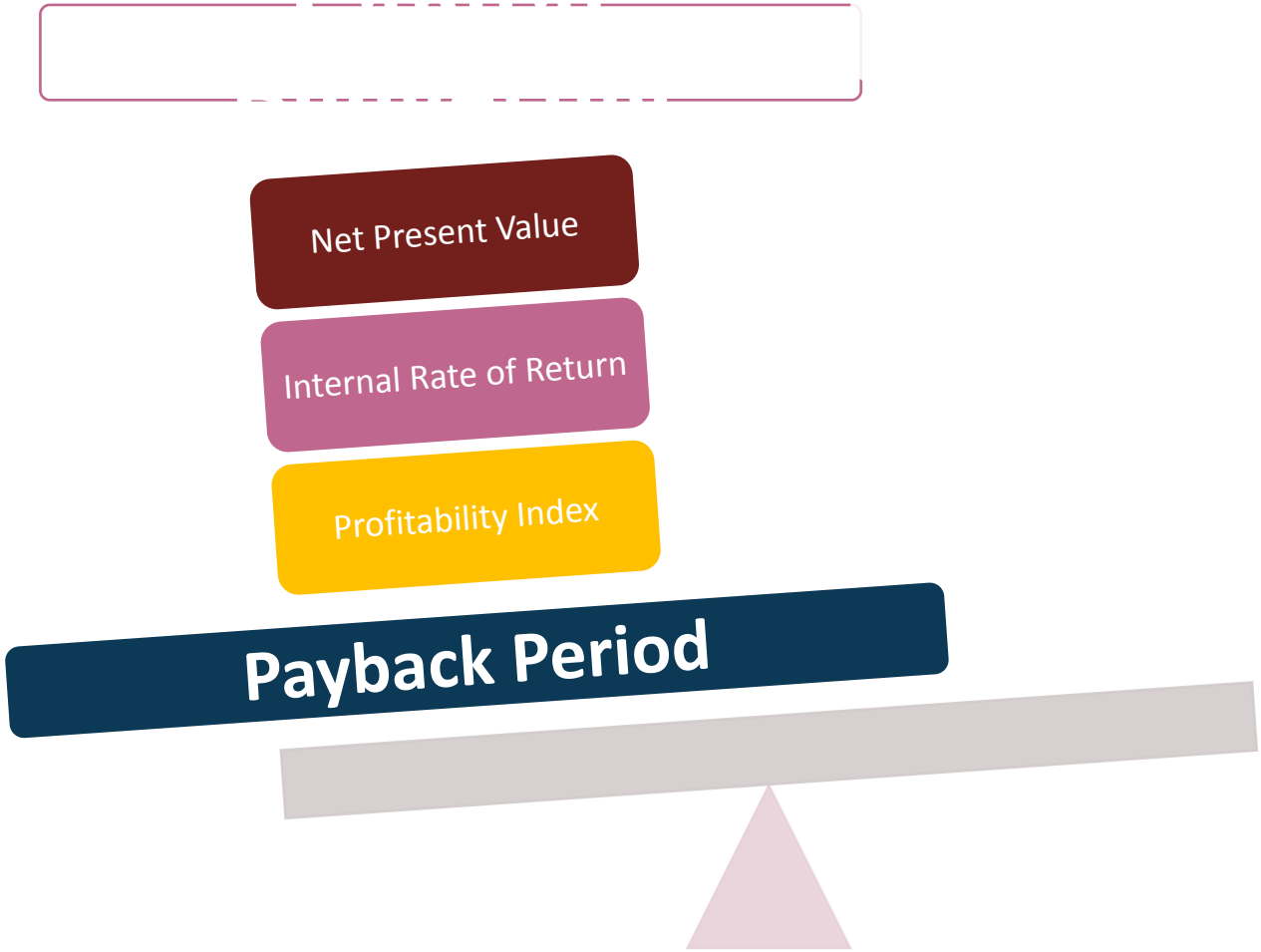
K (Discount rate)	12
K%	0.12
(1+K)	1.12

Calculating the Cash inflows				
Year (n)	FV	$(1+K)^n$		$PV = FV \times \frac{1}{(1+k)^n}$
1	5000	$(1+K)^1$	1.12	**4464.29**
2	9000	$(1+K)^2$	1.254	**7174.74**
3	10000	$(1+K)^3$	1.4049	**7117.80**
4	10000	$(1+K)^4$	1.57352	**6355.18**
5	3000	$(1+K)^5$	1.762342	**1702.28**
Total Cash Inflow				**26814.29**

Profitabilty Index = Cash inflow / Cash Outflow		
Cash Inflow	=	**26814.29**
Cash Outflow	=	**25000**
PI	=	1.07

Profitability Index is 1.07 and since it is greater than 1, we can accept the project.

Payback Period



Payback Period

The time it takes for the project to generate money to pay for itself.

Payback period is the number of years required to recover the cash outflow invested in the project.

The project would be accepted if its payback period is less than the maximum or standard payback period set by Industry, Senior Leadership.

In terms of Projects ranking, it gives highest ranking to the project with the shortest payback period.

Note: In general, the discounted cash flow is not considered for Pay back period. Some do, but most don't!

Payback Period

A sum of $25,000 invested today in an IT project, may give a series of cash inflows in future as described below.

$ 5,000 in year 1
$ 9,000 in year 2
$ 10,000 in each of year 3
$ 10,000 in each of year 4
$ 3,000 in year 5

What is the Payback Period (Non-discounted)?

Year	Cash Inflow	Cumulative Cash flow	
Year1	5000	**5000**	
Year2	9000	**14000** (5000+9000)	
Year3	10000	**24000** (5000+9000+10000)	
Year4	10000	**34000** (5000+9000+10000+10000)	**833.33 Per month**
Year5	3000	**37000** (5000+9000+10000+10000+3000)	

Payback Period (Non-discounted) = In between 3 years 1 month and 3 years 2 months

Important: Few tips to Remember

- ✔ Always choose projects with highest NPV.
- ✔ If NPV is same for the given projects, choose the project with highest IRR.
- ✔ If NPV, IRR remains the same for the given projects, choose the projects with early pay back period.
- ✔ NPV = All Cash Inflows – Cash Outflows
- ✔ PI = All Cash Inflows / Cash Outflows
- ✔ IRR = Discount rate at which the NPV becomes zero, this tell us what is the percent of return for the project.
- ✔ Payback period is a major consideration for every project, business or organization, it tells us how soon we can recover our investment and this investment can be utilized for other business needs/projects later on.

Quick Recap – Concepts Learned

- Opportunity Cost / Discounted Cash flow
- Time Value of Money
- Calculating Future value
- Calculating Present value

Based on the above concepts, we learnt how to solve

- Capital Budgeting techniques.

  ✔ Net Present Value - NPV
  ✔ Internal Rate of Return - IRR
  ✔ Profitability Index – PI
  ✔ Payback Period - PBP